

# Software Requirements Specification (SRS)

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## GalactusAI — AI-Powered Resume Analyser

**Version:** 1.0

**Date:** April 2026

**Live URL:** <https://galactusai.netlify.app/>

## 1. Introduction

### 1.1 Purpose

GalactusAI is an AI-powered web application that analyses a user's resume and provides detailed feedback including ATS score, strengths, weaknesses, missing keywords, and suitable job roles.

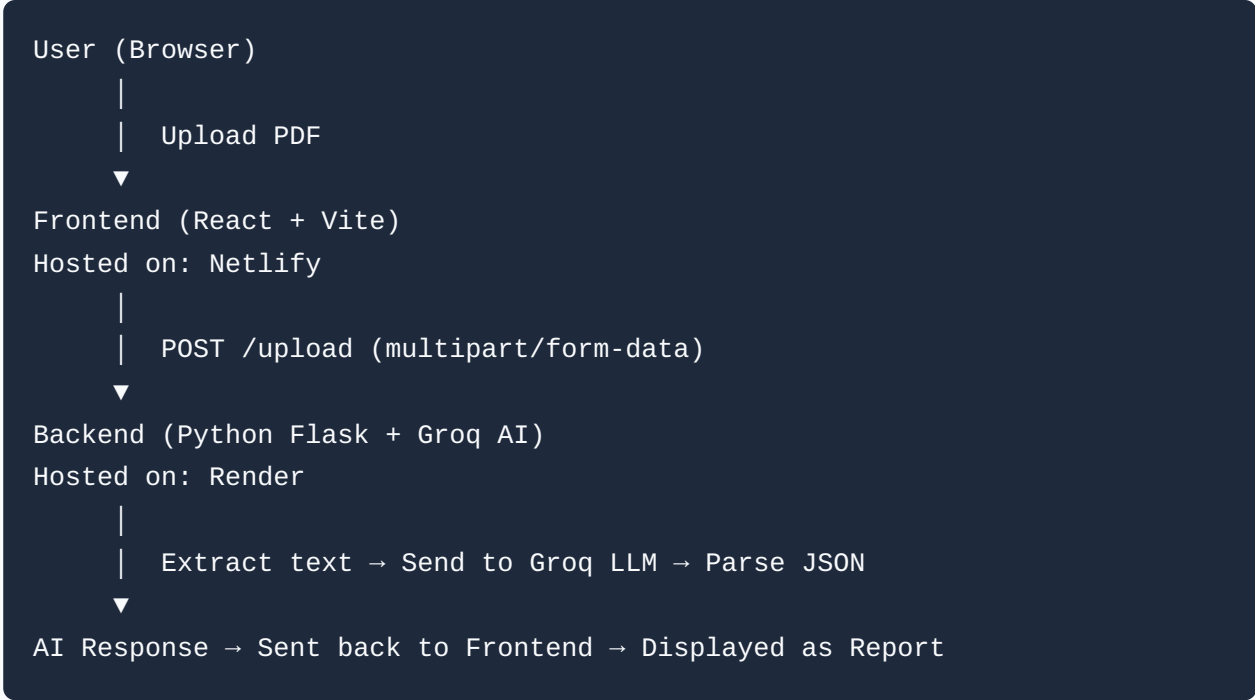
### 1.2 Scope

- Users can upload their resume (PDF format)
- The system extracts text from the PDF
- AI analyses the resume and returns a structured report
- Results are displayed in a visual dashboard

### 1.3 Intended Users

- Students and freshers preparing for job applications
- Working professionals looking to improve their resume
- Anyone who wants ATS-optimized resume feedback

## 2. System Overview



## 3. Functional Requirements

| ID    | Requirement                                                                                  |
|-------|----------------------------------------------------------------------------------------------|
| FR-01 | User can upload a PDF resume (max 5MB)                                                       |
| FR-02 | System extracts text from the uploaded PDF using pdfplumber                                  |
| FR-03 | System sends extracted text to Groq AI (LLaMA 3.3 70B model)                                 |
| FR-04 | AI returns an Overall Score (0–100)                                                          |
| FR-05 | AI returns an ATS Compatibility Score (0–100)                                                |
| FR-06 | AI returns Section-wise scores (Contact, Summary, Experience, Skills, Education, Formatting) |
| FR-07 | AI returns a list of Strengths found in the resume                                           |
| FR-08 | AI returns a list of Improvements needed                                                     |
| FR-09 | AI returns Keywords found and Missing Keywords                                               |

|       |                                                          |
|-------|----------------------------------------------------------|
| FR-10 | AI returns 5 Suitable Job Roles based on the resume      |
| FR-11 | AI returns a one-line Verdict about the resume           |
| FR-12 | User can go back and upload another resume               |
| FR-13 | System shows error if file is not a PDF or is unreadable |

## 4. Non-Functional Requirements

| ID     | Requirement                                                             |
|--------|-------------------------------------------------------------------------|
| NFR-01 | Response time should be under 30 seconds                                |
| NFR-02 | Application should work on both desktop and mobile browsers             |
| NFR-03 | PDF file size limit is 5MB                                              |
| NFR-04 | API keys must not be exposed on the frontend                            |
| NFR-05 | Application should be available 24/7 (Netlify + Render free tier)       |
| NFR-06 | Canvas animation should be smooth — reduced particles on mobile devices |

## 5. Tech Stack

| Layer            | Technology                         |
|------------------|------------------------------------|
| Frontend         | React 18, Vite, Tailwind CSS       |
| Backend          | Python 3, Flask, Flask-CORS        |
| PDF Parsing      | pdfplumber                         |
| AI Model         | Groq API — LLaMA 3.3 70B Versatile |
| Frontend Hosting | Netlify                            |
| Backend Hosting  | Render (Free Tier)                 |

## 6. API Reference

### POST /upload

Accepts a PDF resume file and returns AI analysis.

#### Request:

```
Content-Type: multipart/form-data
Body: resume = <PDF file>
```

#### Response (Success):

```
{
  "success": true,
  "fileName": "my_resume.pdf",
  "analysis": {
    "overallScore": 62,
    "atsScore": 55,
    "sections": {
      "contactInfo": 7,
      "summary": 5,
      "experience": 20,
      "skills": 14,
      "education": 16,
      "formatting": 8
    },
    "strengths": ["..."],
    "improvements": ["..."],
    "keywords": ["..."],
    "missingKeywords": ["..."],
    "suitableRoles": ["..."],
    "verdict": "..."
  }
}
```

#### Response (Error):

```
{  
  "error": "Only PDF files are allowed"  
}
```

## 7. Constraints & Limitations

- Render free tier **sleeps after 15 min of inactivity** — first request may take 30–50 seconds to wake up
- Only **PDF files** are supported (no Word/image resumes)
- **Scanned image PDFs** will fail — text must be selectable in the PDF
- AI scoring is **not 100% deterministic** — results may vary slightly between uploads
- Free tier Groq API has **rate limits** on heavy usage

## 8. Future Enhancements (Possible)

- Support for .docx (Word) resume format
- Side-by-side resume comparison
- Job description matching — paste a JD and get a match score
- Download report as PDF
- User login and resume history
- Cover letter generator